

Deep Learning for Computer Vision

113-1/Fall 2024

<https://cool.ntu.edu.tw/courses/41702> (NTU COOL)

<http://vllab.ee.ntu.edu.tw/dlcv.html> (Public website)

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Deep Learning for Computer Vision

- Time:
 - Tuesday 2, 3, 4
 - ~~09:10-12:20~~ → **09:30-12:20** with 1 short break
- Location:
 - BL-112
- Websites:
 - NTU COOL; primary website: <https://cool.ntu.edu.tw/courses/41702>
 - Public use: <http://vllab.ee.ntu.edu.tw/dlcv.html>

Deep Learning for Computer Vision (cont'd)

- **Required** Knowledge & Skills
 - Knowledge of linear algebra, vector calculus, and probability
 - Machine learning & deep learning (this is **NOT** an entry-level course)
 - Programming skills (in Python)
- *If any of the above skills not available, please come back when ready.*
- This course is offered in **Mandarin** this semester.
Q&As & discussions can be done in any languages communicable.
- To encourage class attendance, lectures will be recorded but **NOT** uploaded to COOL until a week or two after.

What to Expect from this Course?

- (Deep) Learning-Based Computer Vision
 - ~~Fundamentals of machine learning~~ (Sorry, it's required!)
 - **Deep learning** technologies for visual classification, synthesis, and beyond (e.g., LLM & foundation models)
 - Different from courses like *Computer Vision*, *3D Vision*, *Applied Deep Learning*, etc.
- Practical Experiences
 - Assignments and projects dealing with real-world visual data
 - Final projects associated with challenges from CVPR, ICCV, etc.
 - Possible cash prize!!
- Lots of work with fast pace, but hopefully helpful with lots of fun!



FUN!

Disclaimer

- You are responsible for **your own computing resources**.
(e.g., GPUs, Colab Pro for \$11/month, etc.)
You can't finish this course w/o using GPUs.
- While doing my best to stick to what we announce,
syllabus, course policy, or HW/project details might change over time.
(DL/CV/AI are fast growing/changing research topics...)
- Yes, **up to 20% of students** chose to drop the course in previous years.
- And, yes, **we did fail students in the past years (~5% each semester)**.



Required Computing Resources

- If you're using your own machines...(update by HW#1)
 - Ubuntu 20.04.1 LTS
 - NVIDIA GeForce RTX 2080 Ti (11GB)
 - GNU bash, ver. 5.0.17(1)-release

Course Information

- Teaching Team & Office Hours
- Course Policy
- How to enroll in this class if not already in?

About Myself

- **Education**



- **Ph.D./M.S. in Electrical & Computer Engineering** 2002 – 2009
Carnegie Mellon University, Pittsburgh, USA



- **B.S. in Electrical Engineering** 1997 – 2001
National Taiwan University, Taipei, Taiwan

- **Experiences**



- **NVIDIA** AI研究總監, NVIDIA Research 2022/08 – present

- **Professor**, Dept. EE, National Taiwan University



- **Principal AI Consultant**, Inventec 2021 – 2022

AICS



- **AI Advisory Consultant**, ASUS Intel. Cloud Services (AICS) 2019 – 2022

- **Associate Professor**, National Taiwan Univ. 2017 – 2019

- **Deputy Director**, CITI, Academia Sinica 2015 – 2017



- **Associate/Assistant Research Fellow** 2009 – 2017

Research Center for IT Innovation (CITI), Academia Sinica

- **Office Hour**

- After class, or by appointment via email (ycwang@ntu.edu.tw)

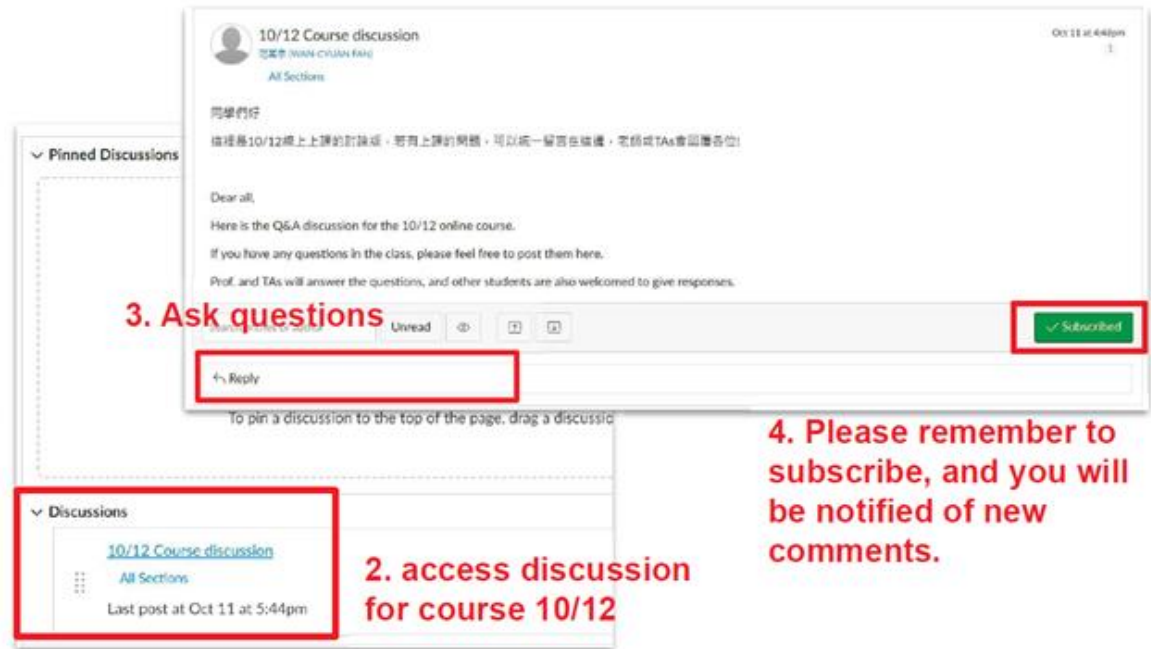
TAs & Office Hours

- We have 8 TAs this semester.
 - Will post and update the TA info on DLCV website
- If any questions, you are encouraged to
 - Ask at NTU COOL (anonymous is fine)
 - Contact your TA **by email at** ntudlcv@gmail.com
- Q&A @ NTU COOL
 - “Discussions” -> “Q&A for Date” -> Ask your questions @ Reply

The screenshot displays the NTU COOL website interface. On the left, a sidebar menu for the course '深度學習於電腦視覺 (COMME5052)' includes links for Home, Syllabus, Modules, Announcements, Assignments, Quizzes, Discussions, and Grades. The 'Discussions' link is highlighted with a red box and labeled '1. click discussions'. The main content area shows a 'Pinned Discussions' section with a link to '10/12 Course discussion' highlighted by a red box and labeled '2. access discussion for course 10/12'. Below this, the discussion page for '10/12 Course discussion' is shown, featuring a 'Reply' button highlighted by a red box and labeled '3. Ask questions'. A green 'Subscribed' button is also highlighted with a red box and labeled '4. Please remember to subscribe, and you will be notified of new comments.'

Q&A @ NTU COOL

- “Discussions” -> “Q&A for Date” -> Ask your questions @ Reply

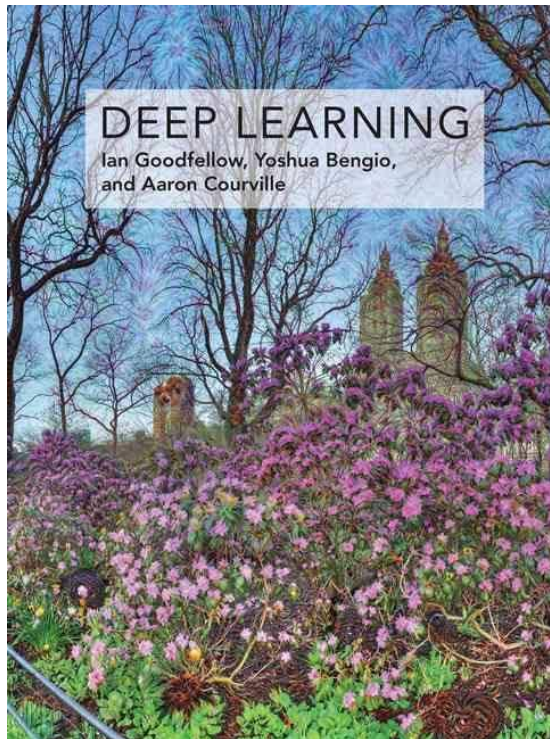


A Very Tight Schedule...

Week	Date	Topic	Course Materials	Remarks
1	09/03	Course Logistics & Registration; Intro to Neural Nets		
2	09/10	Convolutional Neural Networks & Image Segmentation		HW #1 out
3	09/17	No class		Mid-Autumn Festival
4	09/24	Generative Models (I) - AE, VAE & GAN		HW #1 due
5	10/01	Guest Lecture		ECCV week
6	10/8	Generative Models (II) - Diffusion Model		HW # 2 out
7	10/15	Recurrent Neural Networks & Transformer		
8	10/22	Transformer; Vision & Language Models		
9	10/29	Vision & Language Models		HW #2 due; HW #3 out
10	11/05	Multi-Modal Learning; Parameter-Efficient Finetuning		
11	11/12	Unlearning, Debiasing, and Interoperability		
12	11/19	3D Vision		HW #3 due; HW #4 out
13	11/26	Transfer & Adversarial Learning		Final Project Announcement
14	12/03	Federated Learning		HW #4 due
15	12/10	Progress Check for Final Projects		NeurIPS week
17	12/25 Wed	Final Project Presentation		

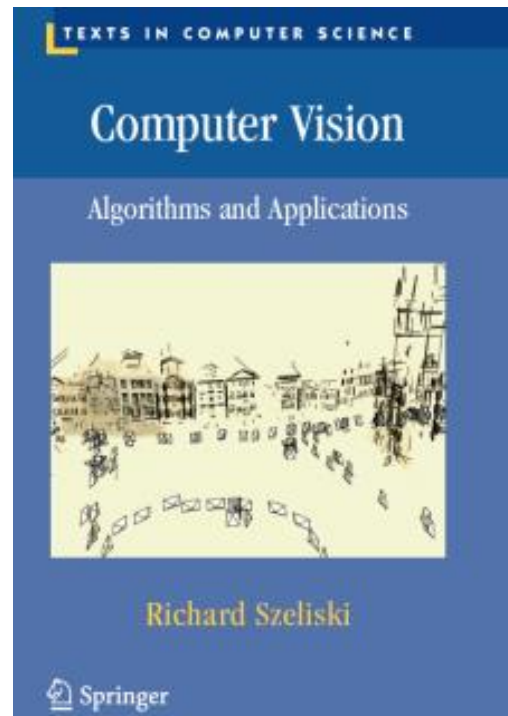
Textbook (Optional)

- Deep Learning, MIT Press
 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville
 - Free online versions available at www.deeplearningbook.org

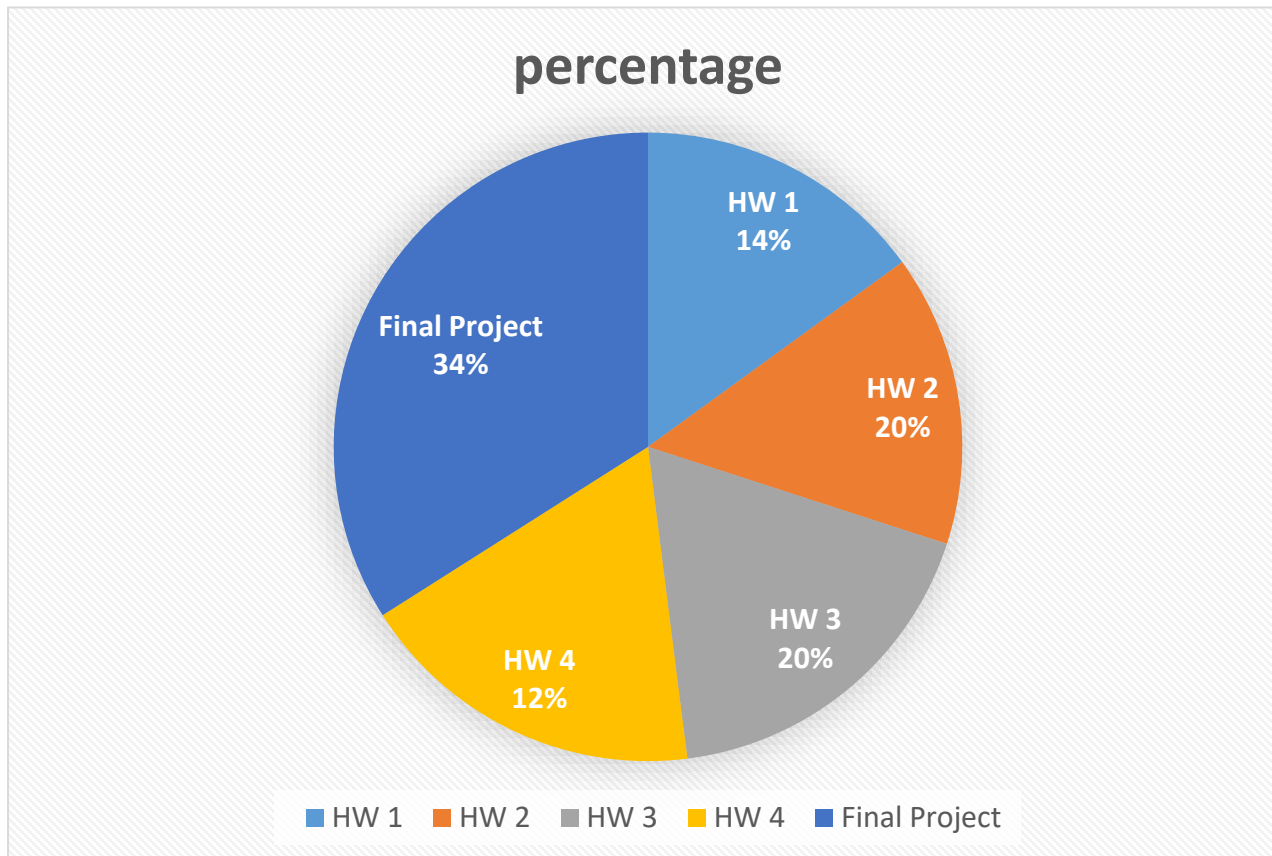


Textbook (Optional)

- Computer Vision: Algorithms and Applications, Springer
 - Richard Szeliski
 - Free online versions available at <http://szeliski.org/Book/>



About Grading



Bonus points possibly available for both HWs & lecture interaction

About Grading (cont'd)

- HW assignments **(66%)**
 - ~~HW 0 (non-DL HW, 0% but required)~~
 - HWs #1~4 (12~20% each with possible bonus points)
 - ~~HW #5 (optional yet with bonus points)~~
- Final project + poster presentation + code, etc. **(34%)**
- **Bonus points**
 - Course participation (e.g., interaction, Q&A, etc.)
 - Extra challenges in HWs
 - Excellent performance for final project (e.g., competitions, publication submissions, etc.)



Final Grade

Letter Grading System	Definition	Grade Points	Conversion Scale
A+	All goals achieved beyond expectation	4.3	90-100
A	All goals achieved	4.0	85-89
A-	All goals achieved, but need some polish	3.7	80-84
B+	Some goals well achieved	3.3	77-79
B	Some goals adequately achieved	3.0	73-76
B- (passing grade for graduate students)	Some goals achieved with minor flaws	2.7	70-72
C+	Minimum goals achieved	2.3	67-69
C	Minimum goals achieved with minor flaws	2.0	63-66
C- (passing grade for undergraduate students)	Minimum goals achieved with major flaws	1.7	60-62
F	Minimum goals not achieved	0	59 and below
X	Not graded due to unexcused absences or other reasons	0	0
W	Withdrawal		
NG	No grade reported		
IP	In progress		
TR	Transfer credit		
EX	Exempted		

About Course HWs/Projects

- About HW late policy
 - We offer free late days (up to **THREE** days)
in case you have dates, midterms, HW dues for other course, etc.
(如期中考、專題、社團、約會、找不到人抄)
 - 1 min ~ 23 hr 59 min all count as ONE late day.
 - After HW due day, a penalty of **30%** per day.
 - We'll maximize your final score
based on HW scores and the late days used.
 - **No** late submission for the final project, obviously.

About Course HWs/Projects



- About Final Project
 - Details to be announced around mid semester.
 - 3~4 people per group
 - Will check your code after presentation (to finalized your scores)
 - Selected topics possibly come with cash prizes.
 - Evaluated by instructor, TAs, and possibly guest judges
 - (Intra/inter-group) peer evaluation will be conducted.
 - Snack/drinks will be provided during final presentation.



Academic Integrity

- Can discuss HW with peers, but DO NOT copy and/or share code
 - Plagiarism is against university policy.
 - Violation in ANY form for HWs & final project would result in F.
 - We gave at least five Fs in previous semesters for the above cases.
- Do not directly use code/results from Internet unless you have permissions.
 - If not sure, ask!
 - If so, do specify in your HW/project.
- No double dipping! **CANNOT** use your research work as your final project.
 - However, you are encouraged to extend your previous work.
 - Also, you are encouraged to turn your high-quality projects into publications.

DOs and DONTs for the TAs (& Instructor)

- Do NOT send **private** messages to the TAs via Facebook, etc.
 - TAs are here to help, but they are not your tutors 24/7.
- TAs will **NOT** debug for you, including coding, environmental, library dependency problems.
- If you cannot make the TA hours, please email & schedule an appointment instead of stopping by the lab directly.
- Obviously, TAs do **NOT** answer questions not related to the course...

How to Sign Up If Not Already In?

- **Capacity**

- Classroom capacity: 120; currently registered: 100
- About **20** students can be additionally added.

- **Priority**

- EECS > Engr. with research needs > Sci. > others
 - Will consider seniority, backgrounds, programming skills, etc.
- If you are still interested in this course and plan to enroll, please fill in the following form **Sept 3rd Tue between 9am-5pm:**
<https://forms.gle/M6Grdrm3F2YsBFeFA>
- We will announce the enrollment results via email no later than **Sept 6th Fri 6pm. All decisions are final.**



Any Questions?

Course registration at <https://forms.gle/M6Grdrm3F2YsBFeFA>
by **Tue Sept. 3rd 5pm!**